



Name : _____

Number : _____

HURLSTONE AGRICULTURAL HIGH SCHOOL

YEAR 12 2009

MATHEMATICS

TRIAL HIGHER SCHOOL CERTIFICATE

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General Instructions

- Reading time : 5 minutes
- **Working time : 3 hours**
- Attempt **all** questions
- **Start a new answer booklet for each question**
- All necessary working should be shown
- This paper contains 10 questions worth 12 marks each. Total Marks: **120 marks**
- Marks may not be awarded for careless or badly arranged work
- Board approved calculators and mathematical templates may be used
- This examination paper must **not** be removed from the examination room

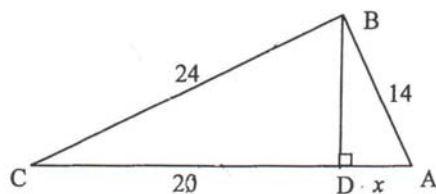
QUESTION 1. *Start a new answer booklet.*

Marks

- | | | |
|----|---|---|
| a) | Factorise $3x^3 + 24$ | 2 |
| b) | Rationalise the denominator and simplify: $\frac{2}{5-\sqrt{3}}$ | 2 |
| c) | Express $\frac{3x+2}{2} - \frac{x-1}{5}$ as a single fraction, in simplest form | 2 |
| d) | Solve the inequality, graphing your solution on a number line: $ 2x-3 < 7$ | 2 |
| e) | Write, in scientific notation, correct to 2 significant figures: $\frac{e^{-3.5}}{4}$ | 2 |
| f) | Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$ | 2 |

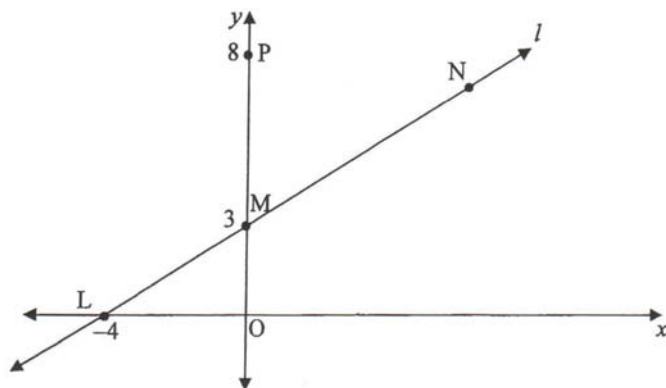
QUESTION 2. Start a new answer booklet.

a)



Find x , the length of AD , as an exact value. Justify your answer.

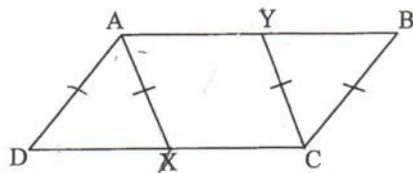
b)



For the diagram shown, M is the midpoint of the interval LN

- Find the coordinates of the point N .
- Show that $\angle NPL$ is a right angle.
- Find the equation of the circle that passes through the points N , P and L .

- c) $ABCD$ is a parallelogram. The point X lies on CD , the point Y lies on AB , and $AX = CY = BC$, as shown in the diagram.



- Explain why $\angle ADX = \angle CBY$.
- Show that $AD = AX$.
- Show that triangles ADX and CBY are congruent.

Marks

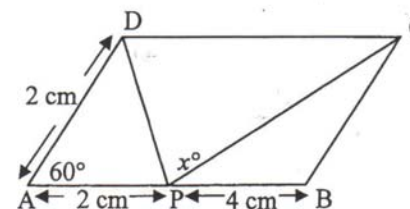
2

QUESTION 3. Start a new answer booklet.

- a) Use the formula $a^2 = b^2 + c^2 - 2bc \cos A$ to find A , correct to the nearest degree, when $a = 22$, $b = 12$, $c = 13$, and A lies between 0° and 180° .

2

b)



NOT TO SCALE

In the figure (not to scale), $ABCD$ is a parallelogram in which $AB = 6$ cm, $AD = 2$ cm, and $\angle DAB = 60^\circ$. The point P on AB is such that $AP = 2$ cm, and $\angle DPC = x^\circ$.

- Find the length of DP , giving reasons.
- Use the cosine rule for each of the triangles PBC , PCD to show that $\cos x^\circ = \frac{-\sqrt{7}}{14}$.

2

3

- c) Find all the values of θ , for $0^\circ \leq \theta \leq 180^\circ$, such that $\cos 2\theta = \frac{1}{2}$.

2

- d) Sketch the graph of:

$$f(x) = \begin{cases} 5 & \text{if } x > 2 \\ x^2 & \text{if } 0 \leq x \leq 2 \\ 2x - 1 & \text{if } x < 0 \end{cases}$$

3

1

1

2

QUESTION 4. Start a new answer booklet.

a) Given that $\log_k 5 = 0.627$ and $\log_k 2 = 0.270$, find the value of:

(i) $\log_k 10$

(ii) $\log_k 25$

b) Differentiate the following with respect to x :

(i) $(e^{2x} + 1)^3$

(ii) $\frac{\log_e x}{x+1}$

c) Find $\int \frac{x}{x^2+2} dx$

d) Evaluate $\int_0^1 (e^{2x} - x) dx$

e) Consider the equation $\log_e y = x \log_e \left(\frac{1}{2}\right)$.

Write an expression for $y = f(x)$.

QUESTION 5. Start a new answer booklet.

Mark:

a) The first three terms of a geometric series are:

$$4a^2b^2, x, a^2 + 2ab + b^2$$

Find x in terms of a and b .

b) A hole in a water reservoir wall will allow through it 50L more for each hour that it remains undetected. At the moment that this hole was detected, water was leaking through it at the rate of 1200L/h.

(i) Write down the first three terms of a series which represents the water lost through the hole for each of the first three hours.

(ii) For how long had the water been leaking when the hole was detected?

(iii) What was the total volume of water lost through the hole, up to the time when it was detected?

c) For the series:

$$1 + \sin A + \sin^2 A + \sin^3 A + \dots$$

(i) Explain why the series has a limiting sum.

(ii) Find the exact value of this limiting sum when $A = \frac{4\pi}{3}$.

d) An investment fund intends to pay interest at the rate of 6% p.a. every six months.

(i) If an investment of \$250 is made today, what amount (ie. principal plus interest) will be available for withdrawal in 10 years time?

(ii) If nineteen further investments of \$250 are made every six months, show that the amount available for withdrawal in ten years time will be \$6919 to the nearest dollar. (Assume that no withdrawals are made from the fund during this time.)

QUESTION 6. Start a new answer booklet.

- a) Consider the following statement:

“In 2008, Fiji had a population of approximately 906 000. The number of fatalities on Fijian roads was 78. If similar figures were to be maintained into the future, the probability of a Fijian, chosen at random, being killed in a road accident in any particular year is approximately 1 in 12 000.”

Comment upon the validity of this statement.

2

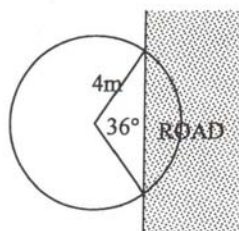
- b) A General Practitioner has compiled statistics on patients visiting his practice. He has determined that during the winter months, there is a 45% chance that a patient will require treatment for influenza and a 15% chance that a patient will require treatment for food poisoning.

- (i) What is the probability that one of the doctor's patients will require treatment for influenza **and** food poisoning during the same season. 1
- (ii) What is the probability that one of the doctor's patients will require treatment for either influenza **or** food poisoning during the same season. 2

- c) A coin is specially weighted so that the probability of tossing a “head” on any single toss is twice that of tossing a “tail”.

- (i) What is the probability of tossing a “head” with this coin. 1
- (ii) Calculate the probability of tossing at least one “tail” if the coin is tossed five times. 1

- d)



A sprinkler spraying water in a circular pattern of radius 3m is watering a lawn adjacent to a straight section of road as shown in the diagram. The angle subtended by the road at the sprinkler head is 36° .

- (i) Convert 36° to radians. Give your answer in terms of π . 1
- (ii) Find the area of road being watered in square metres correct to 2 decimal places. 2
- (iii) Calculate the volume of water being wasted each hour if the sprinkler delivers 3.5kL per hour. Give your answer to the nearest litre. (Assume the sprinkler disperses water evenly over its spray area.) 2

Marks

QUESTION 7. Start a new answer booklet.

a) $\int (2x+3)^3 dx$

1

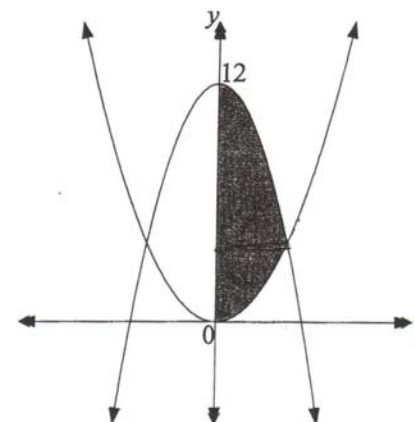
b) Evaluate $\int_1^4 \frac{x^2+8}{x^2} dx$

3

c) Given $f'(x) = 3x^2 + x$, find $f(x)$ given $f(-2) = 4$

2

- d) The graphs of the curves $y = x^2$ and $y = 12 - 2x^2$ are shown in the diagram.



- (i) Find the points of intersection of the curves

1

- (ii) Calculate the area between the two curves

2

- (iii) The shaded region between the curves and the y axis is rotated about the y axis. By splitting the shaded region into two parts, or otherwise, find the volume of the solid formed.

3

QUESTION 8. Start a new answer booklet.

- a) If $A^m = 3$, find the value of $A^{4m} - 5$ 2
- b) Use Simpsons Rule, with three function values, to approximate $\int_3^5 x \log_e x \, dx$ 2
- c) Consider the function $y = xe^{-x}$.
- (i) Find the coordinates of any stationary point(s) and determine their nature 3
- (ii) Describe the behaviour of the function as x :
- increases to negative infinity
 - increases to positive infinity
- 2
- (iii) Does the graph pass through the origin? Justify your answer. 1
- (iv) Sketch the graph of this function 2

QUESTION 9. Start a new answer booklet.

- a) Find the equation of the tangent to the curve $y = \sqrt{x-1}$ at the point $(2,1)$. 2
- b) Solve for x : $x^2 - 8x + 12 > 0$ 2
- c) Point $P(x, y)$ is a point on the parabola $y = x^2$.
- (i) Show that the distance, S , from the line $y = 2x - 5$ is given by $S = \left| \frac{2x - x^2 - 5}{\sqrt{5}} \right|$. 2
- (ii) Show that $f(x) = 2x - x^2 - 5$ is negative definite. 2
- (iii) Hence show that $S = \frac{x^2 - 2x + 5}{\sqrt{5}}$. 1
- (iv) Hence find the shortest distance possible between the point $P(x, y)$ and the line $y = 2x - 5$. 3

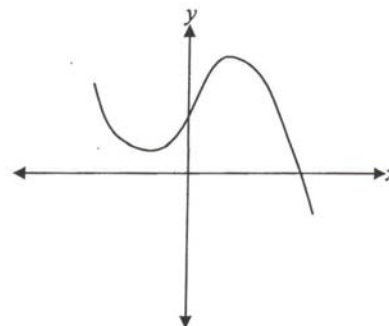
- a) For the parabola $(x-2)^2 = 8(y+1)$, find:

- (i) the focal length 1
- (ii) the co-ordinates of the vertex 1
- (iii) the co-ordinates of the focus 1

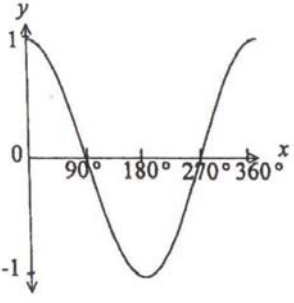
- b) The parabola $y^2 = 4x$ is reflected in the y axis. What is the equation of the resultant parabola formed? 1

- c) Show that the locus of a point, $P(x, y)$, which is always 5 units from the point $A(3, 6)$ is $x^2 - 6x + y^2 - 12y + 20 = 0$. 3

- d) The diagram show a graph of a certain function $y = f(x)$.



- (i) Copy this graph into your writing booklet.
- (ii) On the same set of axes, draw a sketch of the derivative $f'(x)$ of the function. 2
- e) The quadratic equation $x^2 + mx + 10 = 0$ has one root twice the other.
- (i) If one of the roots is α , write expressions for the sum and product of the roots. 1
- (ii) Hence, or otherwise, find the value of m . 2

Year 12 Trial	Mathematics	Examination 2009
Question No.1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
P3 Performs routine arithmetic & algebraic manipulation involving surds & simple rational expressions		
P4 Chooses and applies appropriate arithmetic, algebraic & trigonometric techniques		
P5 Understands the relationship between a function and its graph		
H3 Manipulates algebraic expressions involving logarithmic & exponential functions		
Outcome	Solutions	Marking Guidelines
P3	a) $3x^3 + 24 = 3(x^3 + 8)$ $= 3(x+2)(x^2 - 2x + 4)$	2 marks : factorises correctly twice 1 mark: factorises correctly once
P3	b) $\frac{2}{5-\sqrt{3}} = \frac{2}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}}$ $= \frac{2(5+\sqrt{3})}{22-3}$ $= \frac{5+\sqrt{3}}{11}$	2 mark: finds common denominator and correctly simplifies 1 mark : significant progress towards correct answer
P4	c) $\frac{3x+2}{2} - \frac{x-1}{5} = \frac{5(3x+2) - 2(x-1)}{10}$ $= \frac{15x+10-2x+2}{10}$ $= \frac{13x+12}{10}$	2 marks : correct answer 1 mark : significant progress towards correct answer
P4	d) $ 2x-3 < 7$ $\therefore -7 < 2x-3 < 7$ $\therefore -4 < 2x < 10$ $\therefore -2 < x < 5$ 	2 marks: correct answer 1 mark : partially correct answer
H3	e) $\frac{e^{-3.5}}{4} = 0.007549 \dots$ $= 7.5 \times 10^{-3}$ to 2 significant figures	2 mark : correct answer 1 mark: correct calculation, incorrect rounding
P5	f) $y = \cos x$ 	2 marks: correct answer 1 mark : partially correct answer

Year 12 Trial	Mathematics	Examination 2009
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question H2		
Outcome	Sample Solution	Marking Guidelines
H2	a) $BD^2 + 20^2 = 24^2$ (Pythagoras' Theorem) $BD = \sqrt{176}$ $x^2 + (\sqrt{176})^2 = 14^2$ (Pythagoras' Theorem) $\therefore x = \sqrt{20}$ $= 2\sqrt{5}$	2 mark ~ Correct answer with reasons 1 mark ~ Correct answer without reasons
H2	b) i) $\frac{-4+x}{2} = 0$ $\frac{0+y}{2} = 3$ $x = 4$ $y = 6$ $\therefore N(4, 6)$	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H2	ii) $m_{NP} = \frac{8-6}{0-4} = -\frac{1}{2}$ $m_{PL} = \frac{8-0}{0+4} = 2$ $m_{NP} \times m_{PL} = -\frac{1}{2} \times 2$ $= -1$ $\therefore NP \perp PL$ $\therefore \angle NPL = 90^\circ$	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H2	iii) Circle centre $M(0, 3)$, radius 5 $x^2 + (y-3)^2 = 25$	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H2	c) i) $\angle ADX = \angle CBY$ (opposite $\angle$'s of parallelogram) ii) $AD = BC$ (opposite sides of parallelogram) $AX = BC$ (given) $\therefore AD = AX$	1 mark ~ correct reasons given
H2	iii) $AD = AX \therefore \triangle ADX$ is isosceles $BC = YC \therefore \triangle CBY$ is isosceles $\therefore \angle ADX = \angle AXD$ and $\angle CBY = \angle CYB$ (equal angles in isosceles Δ 's) $\angle ADX = \angle CBY$ (shown in (i)) $AX = BC$ (given) $\angle AXD = \angle CYB$ (equals of $\angle ADX$ and $\angle CBY$) $\therefore \triangle ADX \cong \triangle CBY$ (AAS)	1 mark ~ correct reasons given 2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.

P3 performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities

P4 chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques

P5 understands the concept of a function and the relationship between a function and its graph

P6 relates the derivative of a function to the slope of its graph

Outcome	Solutions	Marking Guidelines
P4	3. a) $a^2 = b^2 + c^2 - 2ab \cos A$ $22^2 = 12^2 + 13^2 - 2(12)(13) \cos A$ $2(12)(13) \cos A = 12^2 + 13^2 - 22^2$ $\cos A = \frac{12^2 + 13^2 - 22^2}{2(12)(13)}$ $\therefore A = 123.2351\dots^\circ \text{ (by calculator)}$ $\therefore A = 123^\circ \text{ to nearest degree}$	1 mark awarded for partial correct solution 2 marks awarded for complete correct solution
P4	b) (i) $DP = 2cm$ ($\triangle DAP$ is equilateral) Or $DP = 2cm$ (using cosine rule)	1 mark for partial correct solution. 2 marks for complete correct solution

(ii)

Now $\angle PBC = 180^\circ - 60^\circ = 120^\circ$ (co-interior to $\angle DAP$; $DA \parallel CB$)

and $BC = AD = 2cm$ (opposite sides in parallelogram)

In $\triangle PBC$:

$$PC^2 = PB^2 + BC^2 - 2(PB)(BC) \cos \angle PBC$$

$$PC^2 = 4^2 + 2^2 - 2(4)(2) \cos 120^\circ$$

$$PC^2 = 16 + 4 - 16 \left(-\frac{1}{2} \right)$$

$$PC^2 = 28$$

$$PC = \sqrt{28}$$

$$PC = 2\sqrt{7}$$

Now $DC = AB = 6cm$ (opposite sides of parallelogram)

In $\triangle PCD$

$$\cos x^\circ = \frac{DP^2 + PC^2 - DC^2}{2 \times DP \times PC}$$

$$\cos x^\circ = \frac{2^2 + (2\sqrt{7})^2 - (6^2)}{2 \times 2 \times 2\sqrt{7}}$$

$$\cos x^\circ = \frac{4 + 28 - 36}{8\sqrt{7}}$$

$$\cos x^\circ = \frac{-4}{8\sqrt{7}}$$

$$\cos x^\circ = \frac{-1}{2\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$$

$$\cos x^\circ = \frac{-\sqrt{7}}{14}$$

1 mark awarded for partial correct solution leading to the correct value of PC .

2 marks awarded for a further correct partial solution using the cosine rule and giving any correct value for $\cos x^\circ$

3 marks awarded for complete correct solution

P3

(c)

$$\cos 2\theta = \frac{1}{2} \text{ for } 0^\circ \leq \theta \leq 180^\circ$$

$$\therefore 2\theta = 60^\circ \text{ or } (360^\circ - 60^\circ) \text{ for } 0^\circ \leq 2\theta \leq 360^\circ$$

$$\therefore 2\theta = 60^\circ \text{ or } 300^\circ$$

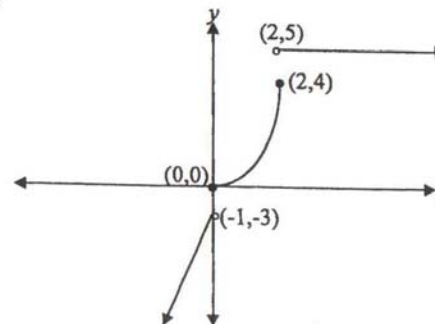
$$\therefore \theta = 30^\circ \text{ or } 150^\circ \text{ for } 0^\circ \leq \theta \leq 180^\circ$$

1 mark awarded for partial correct solution

2 marks awarded for complete correct solution

P5

(d)



1 mark awarded for sketching
 $f(x) = 5$ for $x > 2$

1 mark awarded for sketching
 $f(x) = x^2$ for $0 \leq x \leq 2$

Year 12 Mathematics Trial HSC Examination 2009	
Question No. 4	
Solutions and Marking Guidelines	
Outcomes Addressed in this Question	
H3 manipulates algebraic expressions involving logarithmic and exponential functions	
H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems	
Outcome	Marking Guidelines
(a) (i) H3	$\log_k 10 = \log_k (2 \times 5) = \log_k 2 + \log_k 5 = 0.270 + 0.627 = 0.897$
(ii) H3	$\log_k 25 = \log_k 5^2 = 2 \log_k 5 = 2 \times 0.627 = 1.254$
(b) (i) H3, H5	$\frac{d}{dx} \left((e^{2x} + 1)^3 \right) = 3(e^{2x} + 1)^2 \cdot 2e^{2x} = 6e^{2x} (e^{2x} + 1)^2$
(ii) H3, H5	$\frac{d}{dx} \left(\frac{\log_e x}{x+1} \right) = \frac{(x+1) \cdot \frac{1}{x} - \log_e x \cdot 1}{(x+1)^2} = \frac{x+1 - x \log_e x}{x(x+1)^2}$
(c) H3, H5	$\int \frac{x}{x^2+2} dx = \frac{1}{2} \int \frac{2x}{x^2+2} dx = \frac{1}{2} \log_e (x^2+2) + c$
(d) H3, H5	$\int_0^1 (e^{2x} - x) dx = \left[\frac{1}{2} e^{2x} - \frac{x^2}{2} \right]_0^1 = \frac{1}{2} e^2 - \frac{1}{2} - \left(\frac{1}{2} e^0 - 0 \right) = \frac{1}{2} e^2 - \frac{1}{2} - \left(\frac{1}{2} \right) = \frac{1}{2} e^2 - 1$
(e) H3	$\log_e y = x \log_e \left(\frac{1}{2} \right)$ $\log_e y = \log_e \left(\frac{1}{2} \right)^x$ $\therefore y = \left(\frac{1}{2} \right)^x = 2^{-x}$

Year 12 Mathematics Trial HSC Examination 2009	
Question No. 5	
Solutions and Marking Guidelines	
Outcomes Addressed in this Question	
5 applies appropriate techniques from the study of series to solve problems	
Outcome	Marking Guidelines
H5 (a)	Since the series is geometric: $\frac{T_3}{T_2} = \frac{T_2}{T_1}$ $\text{ie. } \frac{a^2 + 2ab + b^2}{x} = \frac{x}{4a^2b^2}$ $x^2 = 4a^2b^2(a^2 + 2ab + b^2)$ $= 4a^2b^2(a+b)^2$ $x = \pm 2ab(a+b)$
H5 (b) (i)	$T_1 = 50$ $T_2 = 100$ $T_3 = 150$
H5 (ii)	Since the above series is arithmetic: $T_n = 1200 \quad a = 50 \quad d = 50$ $T_n = a + (n-1)d$ $1200 = 50 + (n-1) \cdot 50$ $= 50 + 50n - 50$ $= 50n$ $n = \frac{1200}{50}$ $= 24$ $\therefore$ The water had been leaking for 24 hours when the leak was detected.
H5 (iii)	Total volume of water lost = S_{24} $S_n = \frac{n}{2}(a+l) \quad \text{where } l = T_{24}$ $S_{24} = \frac{24}{2}(50+1200)$ $= 12 \cdot 1250$ $= 15000$ $\therefore$ 15000L of water had been lost when the leak was detected.
H5 (c) (i)	$1 + \sin A + \sin^2 A + \dots$ is a geometric series where $r = \sin A$ Since $-1 \leq \sin A \leq 1$ and provided $\sin A \neq \pm 1$ series will have a limiting sum as limiting sum exists where $-1 < r < 1$

Outcomes Addressed in this Question

H5 applies appropriate techniques from the study of probability and trigonometry to solve problems

H5	(ii)	$r = \sin \frac{4\pi}{3}$ $= -\frac{\sqrt{3}}{2}$ $S_{\infty} = \frac{d}{1-r}$ $= \frac{1}{1+\frac{\sqrt{3}}{2}}$ $= \frac{2}{2+\sqrt{3}}$ $= 4-2\sqrt{3}$	2 marks Correct solution 1 mark Gives correct value for sine ratio OR uses incorrect value in correct formula for limiting sum.
H5	(d) (i)	$A_1 = P(1+r)^n \quad A_1 = P(1+r)^n$ $= 250(1.03)^{20} \quad \text{OR} \quad = 250(1.06)^{10}$ $= 451.53 \quad \quad \quad = 447.71$	1 mark Correct answer. Accept both interest compounded 6 monthly and compounded yearly.
H5		\$447.71 will be available for withdrawal in 10 years.	
H5	(ii)	At the end of 6 months $A_1 = 250(1.03)$ At the end of 12 months (ie. 1 year) $A_2 = 250(1.03)(1.03) + 250(1.03)$ $= 250(1.03)^2 + 250(1.03)$ $= 250(1.03)(1+1.03)$ Similarly, after 3 time periods (18 months) $A_3 = 250(1.03)^3 + 250(1.03)^2 + 250(1.03)$ $= 250(1.03)(1+1.03+1.03^2)$ After 20 time periods ie. 10 years $A_{20} = 250(1.03)^{20} + 250(1.03)^{19} + 250(1.03)^{18} + \dots + 250(1.03)^2 + 250(1.03)$ $= 250(1.03)(1+1.03+1.03^2+\dots+1.03^{19})$ Now, $1+1.03+1.03^2+\dots+1.03^{19}$ is a geometric series with $a=1, r=1.03$ and $n=20$ $S_{20} = \frac{a(r^n-1)}{r-1}$ $= \frac{1.03^{20}-1}{0.03}$ $= 26.8704$ $\therefore A_{20} = 250(1.03)(26.8704)$ $= 6919$ ie. \$6919 is available for withdrawal after 10 years, as required.	2 marks Correct solution 1 mark Substantial progress towards correct solution

Outcome	Solutions	Marking Guidelines
H5	(a) The statement is not valid. It assumes that all Fijians have access to motor vehicles with similar safety levels and the roads being driven on are in a similar state of repair. In fact, many rural dwelling Fijians rarely use motor vehicles, so the probability of them being killed in a road accident is almost zero, whereas a city dwelling Fijian would have a comparably higher probability of being a road fatality. ie. the event of being killed on the road is not equally likely for each Fijian. (b) Probability of requiring treatment for food poisoning $= P(F)$ $= 0.15$ Probability of requiring treatment for influenza $= P(I)$ $= 0.45$	2 marks Correct assessment of validity of statement 1 mark Some aspects of assessment are valid.
H5	(i) $P(F \text{ and } I)$ $= P(F) \times P(I)$ $= 0.15 \times 0.45$ $= 0.0675$	1 mark Correct answer.
H5	(ii) $P(F \text{ or } I)$ $= P(F) + P(I) - P(F \text{ and } I)$ $= 0.15 + 0.45 - 0.0675$ $= 0.5325$ Note: These events are not mutually exclusive	2 marks Correct solution. 1 mark Solution substantially correct.
H5	(c) (i) $P(H) = \frac{2}{3}$	1 mark Correct answer.
H5	(ii) $P(\text{at least 1 tail}) = 1 - P(5 \text{ heads})$ $= 1 - \left(\frac{2}{3}\right)^5$ $= 1 - \frac{32}{243}$ $= \frac{211}{243}$	1 mark Correct solution.
H5	(d) (i) $36^\circ = \frac{36\pi}{180} \text{ radians}$ $= \frac{\pi}{5} \text{ radians}$	1 mark Correct answer.
H5	(ii) $A = \frac{1}{2}r^2(\theta - \sin \theta)$ $= \frac{1}{2} \times 4^2 \left(\frac{\pi}{5} - \sin \frac{\pi}{5} \right)$ $= 0.32$ $\therefore 0.32\text{m}^2$ of the road is being watered.	2 marks Correct solution 1 mark Correct formula and substitution.

H5

(iii)

$$\begin{aligned}
 \text{Water wasted} &= \frac{\text{area of road watered}}{\text{area of circle}} \\
 &= \frac{0.32}{\pi r^2} \times 3.5 \text{ kL} \\
 &= \frac{0.32}{\pi \times 4^2} \times 3.5 \text{ kL} \\
 &\approx 22 \text{ L per hour}
 \end{aligned}$$

2 marks
Correct solution
1 mark
Substantial progress towards correct solution.

Year 12 Trial Question No.7		Mathematics Solutions and Marking Guidelines	Examination 2009
Outcomes Addressed in this Question			
H8 uses techniques of integration to calculate areas and volumes			
Outcome	Solutions	Marking Guidelines	
H8	a) $\int (2x+3)^3 dx = \frac{(2x+3)^4}{4 \times 2} + c$ $= \frac{(2x+3)^4}{8} + c$	1 mark : correct integral	
H8	b) $\int_1^4 \frac{x^2+8}{x^2} dx = \int_1^4 \left(\frac{x^2}{x^2} + \frac{8}{x^2} \right) dx = \int_1^4 (1+8x^{-2}) dx$ $= [x - 8x^{-1}]_1^4$ $= \left[x - \frac{8}{x} \right]_1^4$ $= 4 - 2 - (1 - 8) = 9$	3 marks: correct solution 2 marks: significant progress towards correct solution 1 mark: progress towards correct solution	
H8	c) $f'(x) = 3x^2 + x$ $\therefore f(x) = \frac{3x^3}{3} + \frac{x^2}{2} + c$ $\therefore f(x) = x^3 + \frac{x^2}{2} + c$ $f(-2) = 4, \therefore -8 + 2 + c = 4. \therefore c = 10$ $\therefore f(x) = x^3 + \frac{x^2}{2} + 10$	2 marks : correct answer with justification 1 mark: one of above	
H8	d) (i) $y = x^2$ and $y = 12 - 2x^2$ meet when $x^2 = 12 - 2x^2$. $\therefore 3x^2 = 12$ $\therefore x^2 = 4 \quad x = \pm 2$ $\therefore$ meet at $(-2, 4)$ and $(2, 4)$	1 mark : correct answer	
H8	(ii) Area = Area under top curve - area under bottom curve $= \int_{-2}^2 (12 - 2x^2 - x^2) dx = 2 \int_0^2 (12 - 3x^2) dx$ $= 2 [12x - x^3]_0^2$ $= 2(24 - 8) = 32 \text{ units}^2$	2 marks : correct answer with justification 1 mark : significant progress towards correct answer	
H8	(iii) Volume, in relation to y axis $= \pi \int_a^b x^2 dy$ V = sum of the volume when the area between $y = x^2$ and the y axis between $y = 0$ and $y = 4$ is rotated about the y axis, and the volume when the area between $y = 12 - 2x^2$ and the y axis between $y = 4$ and $y = 12$ rotated about the y axis	3 marks: correct solution 2 marks: significant progress towards correct solution 1 mark: progress towards correct solution	

From $y = x^2$, ie. $x^2 = y$

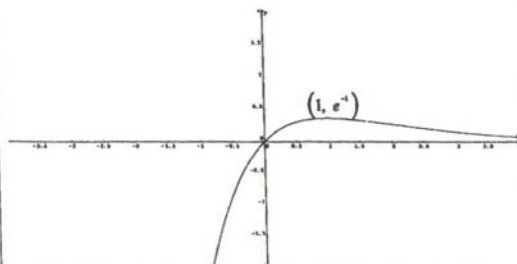
From $y = 12 - 2x^2$, $2x^2 = 12 - y$, $\therefore x^2 = 6 - \frac{y}{2}$.

$$\therefore V = \pi \int_0^4 y dy + \pi \int_4^{12} \left(6 - \frac{y}{2}\right) dy$$

$$\therefore V = \pi \left[\frac{y^2}{2} \right]_0^4 + \pi \left[6y - \frac{y^2}{4} \right]_4^{12}$$

$$\therefore V = 8\pi + \pi(72 - 36 - (24 - 4))$$

$$\therefore V = 24\pi \text{ units}^3$$

12 Mathematics Trial HSC 2009	
Question No. 8 Solutions and Marking Guidelines	
Outcomes Addressed in this Question	
I3	manipulates algebraic expressions involving logarithmic and exponential functions
I5	applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems
Outcome	Solutions
(a) H3	$A^{4m} - 5 = (A^m)^4 - 5$ $= 3^4 - 5$ $= 76$
(b) H3, H5	$\int_3^5 x \log_e x \, dx \approx \frac{1}{3} (3 \log_e 3 + 5 \log_e 5 + 4 \times 4 \log_e 4)$ $= 11.17457874$
(c) (i) H3, H5	$y = xe^{-x}$ $\frac{dy}{dx} = x \cdot -e^{-x} + e^{-x} \cdot 1 = e^{-x}(1 - x)$ $\frac{d^2y}{dx^2} = e^{-x}(-1) + (1 - x) \cdot -e^{-x} = e^{-x}(x - 2)$ Stationary point(s) occur @ $\frac{dy}{dx} = 0$ $e^{-x}(1 - x) = 0$ $\therefore 1 - x = 0 \quad (\because e^{-x} \neq 0)$ $\therefore x = 1$ Test $x = 1$ $\frac{d^2y}{dx^2} = e^{-1}(1 - 2) = -e^{-1} < 0$ $\therefore$ Relative maximum turning point @ $(1, e^{-1})$
(ii) H3	$\lim_{x \rightarrow 0^+} (xe^{-x}) = 0 \text{ or function approaches } y = 0.$ $\lim_{x \rightarrow -\infty} (xe^{-x}) = -\infty \text{ or function gets very big and negative.}$
(iii) H3, H5	When $x = 0$, $y = 0e^{-0} = 0$ $\therefore$ Graph passes through the origin.
(iv) H3, H5	 Award 2 for correct graph, showing relevant details from (i), (ii) and (iii). Award 1 for correct graph, but lacking sufficient detail.

Outcomes Addressed in this Question H5

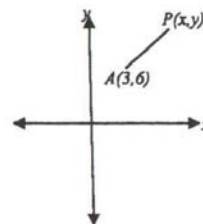
Outcome	Sample Solution	Marking Guidelines
H5	a) $y = \sqrt{x-1}$ $y = (x-1)^{\frac{1}{2}}$ $y' = \frac{1}{2}(x-1)^{-\frac{1}{2}} \cdot 1$ $y' = \frac{1}{2\sqrt{x-1}}$ when $x=2$ $y' = \frac{1}{2}$ eqn of tangent: $y-1 = \frac{1}{2}(x-2)$ $y = \frac{1}{2}x \text{ or } x-2y=0$	2 mark ~ Correct equation 1 mark ~ Substantial progress towards correct solution
H5	b) $x^2 - 8x + 12 > 0$ $(x-6)(x-2) > 0$ $\therefore x > 6, x < 2$	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H5	c) i) $S = \left \frac{2x - x^2 - 5}{\sqrt{2^2 + (-1)^2}} \right $ $= \left \frac{2x - x^2 - 5}{\sqrt{5}} \right $	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H5	ii) $f(x) = 2x - x^2 - 5$ $= -x^2 + 2x - 5$ For negative definite, $a < 0$ and $\Delta < 0$ $a = -1 \quad \Delta = 4 - 4 \times (-1) \times (-5)$ $= 4 - 20$ $= -16$ $\therefore f(x)$ is negative definite.	2 marks ~ Correct solution 1 mark ~ Substantial progress towards correct solution.
H5	iii) since $f(x)$ is negative definite, $f(x) = -f(x)$ $\therefore 2x - x^2 - 5 = -(2x - x^2 - 5)$ $= x^2 - 2x + 5$ $\therefore S = \frac{x^2 - 2x + 5}{\sqrt{5}}$	1 mark ~ correct reasons given
H5	iv) $S = \frac{x^2 - 2x + 5}{\sqrt{5}}$ $S' = \frac{2x - 2}{\sqrt{5}}$ $S'' = \frac{2}{\sqrt{5}}$ stat. pt $S' = 0 \quad \frac{2x-2}{\sqrt{5}} = 0$ $x = 1$ when $x=1$ $S'' = \frac{2}{\sqrt{5}} > 0 \therefore$ Minimum at $x=1$ when $x=1$ $S = \frac{1^2 - 2 + 5}{\sqrt{5}} = \frac{4}{\sqrt{5}}$ $\therefore$ Shortest possible distance is $\frac{4}{\sqrt{5}}$ units.	3 marks ~ Correct solution 2 marks ~ Substantial progress towards correct solution. 1 mark ~ Some progress towards correct solution.

Outcomes Addressed in this Question

P3 performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities

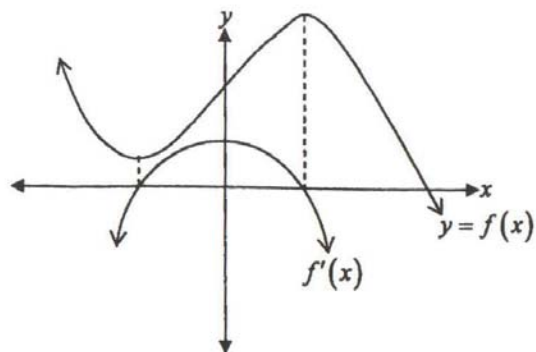
P5 understands the concept of a function and the relationship between a function and its graph

P6 relates the derivative of a function to the slope of its graph

Outcome	Solutions	Marking Guidelines
P5	10. (a) (i) $4a = 8$ $\therefore a = 2$ $\therefore \text{focal length is } 2$ (ii) Vertex (2,-1) (iii) Focus (2,1)	1 mark awarded for correct answer 1 mark awarded for correct answer 1 mark awarded for correct answer
P5	(b) $y = -4x$	1 mark awarded for correct answer
P5	(c)  $PA = \sqrt{(x-3)^2 + (y-6)^2}$ condition is $PA = 5$ $5 = \sqrt{(x-3)^2 + (y-6)^2}$ $\therefore (x-3)^2 + (y-6)^2 = 25$ $\therefore x^2 - 6x + 9 + y^2 - 12y + 36 = 25$ $\therefore x^2 - 6x + y^2 - 12y + 20 = 0$ The locus is $x^2 - 6x + y^2 - 12y + 20 = 0$	1 mark awarded for partial correct solution 2 marks awarded for a further correct partial solution 3 marks awarded for complete correct solution

P6

(d)



1 mark awarded for
partial correct graph

2 marks awarded for
complete correct graph

P3

(e)

(i)

Roots are $\alpha, 2\alpha$

Sum of roots: $3\alpha = -m$

Product of roots: $2\alpha^2 = 10$

1 mark awarded for
correct solution

(ii)

$$3\alpha = -m \dots\dots\dots (A)$$

$$2\alpha^2 = 10 \dots\dots\dots (B)$$

From (A):

$$\alpha = \frac{-m}{3} \dots\dots\dots (C)$$

substitute (C) into (B):

$$2\left(\frac{-m}{3}\right)^2 = 10$$

$$\frac{2m^2}{9} = 10$$

$$m^2 = 45$$

$$m = \pm\sqrt{45}$$

$$m = \pm 3\sqrt{5}$$

1 mark for partial correct
solution.

2 marks for complete
correct solution